

HYDROGEOLOGICAL ZONES OF NEPAL

Comprehensive Engineering Module

Syllabus Coverage: Chapter 9

Prepared For:

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Formatting Note:

10pt Font, Narrow Margins, Continuous Flow.

Date: January 13, 2026

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1 9.1 Hydrogeological Zones of Nepal

1.1 Introduction

The groundwater potential of any region is strictly dictated by its geology. Nepal, with its extreme topographical variation from the low-lying plains of the Terai (60m asl) to the highest peaks in the world (8848m asl), exhibits equally diverse hydrogeological characteristics. The country is vertically divided into three distinct hydrogeological zones: the Terai Zone, the Hilly Zone, and the Mountain Zone.

1.2 1. The Terai Zone (Southern Plains)

The Terai is a distinct hydrogeological unit representing the northernmost extension of the vast Indo-Gangetic plain.

- **Geology:** It consists of thick, unconsolidated alluvial deposits. These sedimentssand, gravel, silt, and clayhave been eroded from the Himalayas and deposited by major rivers (Koshi, Gandaki, Karnali) over millions of years. The sediment thickness can exceed 1500 meters in some places.
- **Potential:** This is the most productive groundwater zone in Nepal, often referred to as the "Groundwater Pow-erhouse." The high porosity and permeability of the alluvial sands create optimal conditions for groundwater storage and transmission.
- **Aquifers:** The region contains rich, regionally continuous aquifer systems. In many areas, multiple aquifer layers exist separated by clay aquitards, allowing for both shallow and deep groundwater development for heavy irrigation and municipal supply.

1.3 2. The Hilly Zone (Mid-Hills & Valleys)

This zone covers the Mahabharat and Siwalik ranges.

- **Geology:** It is composed primarily of consolidated metamorphic and sedimentary rocks such as schist, phyllite, quartzite, and limestone.
- **Potential:** Generally, these hard rocks have very low primary porosity. Groundwater potential is limited and occurs primarily in "secondary porosity"fractures, faults, joints, and weathered zones.
- **Intermontane Valleys:** The exception to the low potential is the presence of Inner Valleys (Dun Valleys) like Kathmandu, Dang, and Pokhara. These valleys act as sediment traps, filled with thick lacustrine (lake) or fluvial (river) deposits that form localized but highly productive aquifers.
- **Springs:** The rural population in the hills relies heavily on natural springs (*muls*). These occur where the water table intersects the ground surface, often at the contact point between permeable fractured rock and impermeable slate/clay.

1.4 3. The Mountain Zone (Himalayas)

- **Geology:** This zone consists of hard crystalline rocks (granite, gneiss) with steep topographical slopes.
- **Potential:** The groundwater potential is very poor. The steep slopes promote rapid surface runoff, leaving little time for infiltration. Furthermore, the lack of soil cover and the non-porous nature of the crystalline bedrock mean there is no storage space for water. Most water resources here are locked in the solid phase (glaciers and snow).

2 9.2 Aquifer System of the Terai

The Terai belt is not uniform. Hydrogeologically, it is stratified into three distinct bands running parallel from North (foothills) to South (Indian border). Understanding this "Schematic Zonation" is critical for engineering design.

2.1 Zone 1: The Bhabar Zone (Recharge Zone)

- **Location:** This is the northernmost strip, approximately 8–12 km wide, lying directly at the foot of the Siwalik (Chure) hills. It represents the "alluvial fan" deposits of rivers entering the plains.
- **Lithology:** The material is extremely coarse, consisting of large boulders, cobbles, pebbles, and coarse sand. Fine sediments (clay/silt) are largely absent as the fast-flowing rivers wash them further south.
- **Hydrogeological Characteristics:**
 - **High Permeability:** Due to the large pore spaces between boulders, water infiltrates this zone very rapidly.
 - **Deep Water Table:** Because the material is so porous, water drains down quickly. The water table is often very deep (>30m), making manual hand pumps infeasible.
 - **Function:** It acts as the primary **Recharge Area** for the entire Terai groundwater system. Monsoon rains sink into the Bhabar zone and travel south underground to pressurize the aquifers in the Southern Terai.
- **Development:** Drilling is technically challenging and expensive due to the presence of large boulders which damage rotary drill bits. Percussion (cable tool) drilling is often required.

2.2 Zone 2: The Middle Terai (Marshy/Transition Zone)

- **Location:** A narrow belt immediately south of the Bhabar zone.
- **Lithology:** A transition mix of sand, gravel, and increasing amounts of silt and clay.
- **Hydrogeological Characteristics:**
 - As the ground slope decreases significantly compared to the Bhabar zone, the velocity of groundwater flow slows down.
 - **Spring Line:** Historically, this was a marshy zone. The groundwater that sank deep in the Bhabar zone would encounter layers of fine sediment here and be forced back to the surface, creating a line of natural springs and wetlands.
 - The water table is generally shallow.

2.3 Zone 3: The Southern Terai (Gangetic Plain)

- **Location:** The southernmost flat plains extending to the Indian border.
- **Lithology:** The sediments are much finer, dominated by layers of fine sand, silt, and clay, interbedded with occasional layers of gravel.
- **Hydrogeological Characteristics:**
 - **Confined Aquifers:** The layers of thick clay act as confining beds (aquicludes/aquitards), separating the water-bearing sand layers.
 - **Artesian Potential:** Because the recharge area (Bhabar) is at a significantly higher elevation (hydraulic head), the deep aquifers in the south are under high hydrostatic pressure. When a tube well punctures the overlying clay, water rises in the pipe. In many locations, the pressure is sufficient for the water to flow out naturally at the surface (Free-flowing Artesian Wells).
- **Development:** This zone is highly suitable for both Shallow Tube Wells (STW) in the upper unconfined layers and Deep Tube Wells (DTW) in the lower confined layers.

Past Exam Questions

Past Exam Questions:

- **2072 Magh (Back) [6 Marks]:** Discuss about the groundwater development possibilities of Terai belt.
Tip: Describe the three zones (Bhabar, Middle, Southern) and emphasize the artesian potential in the south versus the recharge function of the north.

- **2078 Chaitra [4 Marks]:** Discuss about the overall condition of exploitation of groundwater in urban area with reference to Butwal...

Note: Butwal lies near the critical interface of the Bhabar and Middle Terai.

3 9.3 Groundwater Development Possibilities of Terai Belt

The Terai region is the "Granary of Nepal" and holds the country's largest and most productive groundwater reserves. The flat topography and thick alluvial deposits make it highly suitable for large-scale extraction.

3.1 1. Development Potential by Zone

3.1.1 Shallow Tube Wells (STW)

- **Area:** Highly feasible in the Middle and Southern Terai where the water table is shallow (typically 2–5 meters below ground level).
- **Technology:** Simple boring methods (sludging/donkey method) can be used. Centrifugal pumps (suction lift) are effective.
- **Usage:** These are low-cost installations widely used by individual farmers for small-scale irrigation. The vast majority of groundwater irrigation acreage in Nepal is fed by STWs.

3.1.2 Deep Tube Wells (DTW)

- **Area:** Necessary in the **Bhabar Zone** (where the water table is too deep for suction pumps) and for municipal supply in urban clusters of the Southern Terai.
- **Technology:** Requires heavy drilling rigs (Rotary or Percussion) and submersible pumps.
- **Artesian Advantage:** In the Southern Terai, the confined nature of deep aquifers allows for the construction of free-flowing wells, which eliminates energy costs for pumping a massive economic advantage for irrigation projects.

3.2 2. Conjunctive Use Strategy

The immense groundwater potential of the Terai allows for **Conjunctive Use** planning.

- **Wet Season (Monsoon):** Farmers utilize surface water from rivers and diversion canals, which are abundant during rains.
- **Dry Season (Winter/Spring):** When river flow diminishes, farmers switch to groundwater (tube wells).
- **Benefit:** This strategy ensures year-round irrigation, enabling a cropping intensity of greater than 200% (up to 3 crops per year).

3.3 3. Issues & Challenges

- **Arsenic Contamination:** A major hydro-chemical issue in districts like Nawalparasi, Rautahat, and Kailali. Naturally occurring arsenic in the sediment is mobilized into the groundwater (often due to reductive dissolution of iron oxides), rendering shallow aquifers unsafe for drinking (>50 ppb). Deep aquifers are usually safe.
- **Declining Levels:** In dense urban industrial clusters (like Biratnagar, Bhairahawa, Butwal), unrestricted extraction is causing localized declines in the water table.

Past Exam Questions**Past Exam Questions:**

- **2072 Magh (Back) [6 Marks]:** Discuss about the groundwater development possibilities of Terai belt.

4 9.4 Aquifer System of Kathmandu Valley

4.1 Geological Context

The Kathmandu Valley is a bowl-shaped intermontane basin filled with thick layers of fluvio-lacustrine sediments (sand, silt, clay, and gravel) deposited over millions of years. These sediments reach depths of over 500 meters in the central part of the valley. The defining hydrogeological feature is the massive, impermeable layer of black clay, known as the **Kalimati Formation** (or Kalimati Clay), which was deposited when the valley was a paleo-lake.

4.2 Structure of the Aquifer System

While hydrogeologists divide the valley into distinct districts (Northern, Central, Southern), for engineering purposes, the system is best described by its vertical stratification:

4.2.1 A. The Aquiclude (Kalimati Clay)

- **Description:** A thick, cohesive layer of black clay rich in organic matter (peat/methane).
- **Function:** It acts as a massive hydraulic barrier (Aquiclude) separating the surface from the deep underground. It prevents direct vertical recharge of the deep aquifers from rainfall in the city center.

4.2.2 B. Shallow Aquifer

- **Depth:** Found at shallow depths (0 to 30 meters) above the clay layer.
- **Nature:** Mostly Unconfined (water table aquifer).
- **Recharge:** Recharge is direct from rainfall and surface infiltration.
- **Quality:** It is highly vulnerable to anthropogenic pollution. Leaking sewers and septic tanks have contaminated this layer with high levels of Nitrates, Ammonia, and E. Coli bacteria.
- **Usage:** Traditional stone spouts (*dhunge daras*) and shallow dug wells tap this layer. Many are now drying up or are too polluted for use.

4.2.3 C. Deep Aquifer

- **Depth:** Found below the Kalimati clay layer (typically >50 meters, extending to 300+ meters).
- **Nature:** Confined Aquifer. The water is under pressure, trapped between the bedrock below and the Kalimati clay above.
- **Recharge mechanism:** Since the clay blocks vertical rain infiltration, this aquifer is recharged only from the **Northern Foothills** (Tokha, Budhanilkantha, Gokarna). In these northern edges, the clay layer is absent, and coarse sandy deposits are exposed to the surface, allowing rain to sink and flow laterally under the city.
- **Quality:** Generally free from biological pathogens (due to the clay seal) but chemically rich in minerals. It often contains high levels of dissolved Iron, Ammonia, and Methane gas (from the organic clay).
- **Usage:** Heavily exploited by hotels, industries, and the water utility (KUKL) via deep tube wells.

4.3 Schematic Zonation Summary

Feature	Shallow Aquifer	Deep Aquifer
Depth	0 – 30 meters	> 50 meters
Type	Unconfined (Water Table)	Confined (Artesian pressure)
Barrier	None (Directly below surface)	Capped by thick Black Clay (Kalimati)
Recharge	Local Rainfall (Monsoon)	Lateral flow from Northern Hills (Slow)
Quality	Biologically unsafe (Polluted)	High Iron/ Ammonia, Biologically safe
Status	Sustainable (annual refill)	Mining (fossil water), declining fast

Table 1: Comparison of Shallow and Deep Aquifers in Kathmandu

Past Exam Questions

Past Exam Questions:

- **2081 Final [4 Marks]:** Explain in brief the aquifer system of Kathmandu valley.
- **2073 Bhadra [4 Marks]:** Discuss the aquifer system of Kathmandu Valley.
- **2072 Ashwin [6 Marks]:** Discuss about aquifer system of Kathmandu Valley of Nepal.
- **2071 Bhadra [6 Marks]:** Discuss with the help of schematic zones the availability and possibilities of ground-water resources in Kathmandu Valley.

5 9.5 Urban Groundwater Issues

Rapid urbanization in Nepal's major cities (Kathmandu, Butwal, Biratnagar) has led to a massive increase in water demand. With surface water supplies often insufficient or polluted, cities rely heavily on groundwater, creating critical sustainability issues.

5.1 1. Over-Exploitation and Depletion

Over-exploitation is defined as the condition where the average extraction rate exceeds the average natural recharge rate over a prolonged period.

- **Declining Water Levels:** In Kathmandu, the deep aquifer water level is dropping by approximately 2 to 4 meters per year. This indicates "Groundwater Mining" extracting fossil water that is not being replaced.
- **Land Subsidence:** As water is pumped out of the pore spaces in the clay/sand mix, the hydraulic pressure drops. The weight of the soil above compresses the dewatered clay layers, leading to the sinking of the ground surface. This poses a major structural risk to infrastructure in Kathmandu.
- **Drying of Traditional Sources:** As the regional water table drops, it falls below the intake levels of traditional stone spouts and dug wells, destroying centuries-old heritage water systems.

5.2 2. Cone of Depression (Case Studies)

A "Cone of Depression" is the funnel-shaped lowering of the water table around a pumping well. In urban areas, the interference of thousands of wells creates a regional depression.

5.2.1 Case A: Kathmandu Valley

- **Situation:** The valley is a closed "bowl" with limited recharge.
- **The Cone:** Extensive pumping by hotels and industries in the city core has created a deep regional cone of depression.

- **Result:** The hydraulic gradient has reversed in some places. Instead of water flowing from north to south, it flows towards the city center "hole," potentially pulling polluted shallow water down into the deep aquifer if the clay barrier is breached (e.g., by old, unsealed wells).

5.2.2 Case B: Butwal (Terai Urban Center)

- **Situation:** Butwal lies in the Bhabar-Middle Terai transition zone, which is naturally a high-recharge area.
- **Issue:** Despite high recharge, the rapid industrialization and population growth have led to a dense clustering of high-capacity tube wells.
- **Consequence:** The interference between these closely spaced wells creates localized, deep cones of depression. This causes nearby shallow hand pumps used by poorer communities to run dry, creating social inequity.

5.3 3. Groundwater Pollution

- **Shallow Aquifers:** Due to the lack of centralized sewage systems, septic tanks and pit latrines leak directly into the ground. Shallow groundwater in urban Nepal often exceeds safe limits for Nitrates and Coliform bacteria.
- **Deep Aquifers:** While generally protected by clay, they face "Induced Pollution." Over-pumping lowers the pressure in the deep aquifer, creating a suction force that can pull polluted shallow water down through leaks in the clay or poorly constructed well casings.

Past Exam Questions

Past Exam Questions:

- **2078 Chaitra [4 Marks]:** Discuss about the overall condition of exploitation of groundwater in urban area with reference to Butwal and its mitigation measures.

6 9.6 Mitigation & Management

As urban centers face groundwater depletion, the focus has shifted from simple "exploitation" to "management." The goal is to adhere to the principle of **Sustainable Yield**.

6.1 1. Artificial Recharge

Since natural recharge is often blocked by urbanization (concrete sealing the ground) or geological barriers (clay), we must "artificially" inject water back into the aquifers.

- **Rainwater Harvesting (RWH):**
 - *Rooftop RWH:* Collecting rain from building roofs and directing it into Recharge Pits or Dug Wells rather than sewer drains.
 - *Legislation:* It is now mandatory for new large houses in Kathmandu to have recharge pits. This helps bypass the impermeable surface layer.
- **Recharge Ponds (Pokharis):** Reviving traditional ponds (like those in Patan/Bhaktapur) allows standing water to slowly infiltrate and recharge the shallow aquifer.
- **Injection Wells:** Using existing tube wells to pump treated surplus surface water (e.g., from the Melamchi project during the wet season) directly into the deep aquifers under pressure.

6.2 2. Management Strategies (Regulatory Measures)

- **Zoning & Licensing:** Declaring "Critical Zones" (e.g., central Kathmandu) where no new deep tube wells are permitted. Industrial extraction must be licensed and metered.

- **Relocation of Well Fields:** Moving heavy pumping centers from the city core (where clay blocks recharge) to the **Northern Foothills** (Tokha, Gokarna). Here, the aquifer is unconfined and recharges easily; extraction here captures water before it flows under the impermeable clay.
- **Conjunctive Use:** Mandating that industries use surface water or municipal supply during the monsoon and restrict groundwater pumping to the dry season only.
- **Monitoring:** Establishing a network of piezometers to track water levels and quality trends, allowing for data-driven decisions.

Past Exam Questions

Past Exam Questions:

- **2078 Chaitra:** "...and its mitigation measures." [Part of 4 mark question]
Answer: List Artificial Recharge (Rainwater harvesting) and Regulation (Licensing) as key measures.